- **Feathery Appendages:** Seeds or fruits may have feathery appendages which greatly increase their buoyancy to disperse to high altitudes. Examples: *Vernonia* and *Asclepias.*
- **Censor mechanisms:** The fruits of many plants open in such a way that the seeds can escape only when the fruit is violently shaken by a strong wind. Examples: *Aristolochia* and Poppy.

Guess!! Who am I…….? I am dispersed by ant and I have caruncle.

6.4.2 Dispersal by Water (Hydrochory)

Dispersal of seeds and fruits by water usually occurs in those plants which grow in or near water bodies . Adaptation of hydrochory are

- Obconical receptacle with prominent air spaces. Example: *Nelumbo*.
- Presence of fibrous mesocarp and light pericarp. Example: Coconut.
- Seeds are light, small, provided with aril which encloses air.Example: *Nymphaea*.
- The fruit may be inflated. Examples: *Heritiera littoralis*.
- Seeds by themselves would not float may be carried by water current. Example: Coconut.



Figure 6.30: *Nelumbo*



Figure 6.31: Coconut

6.4.3 Dispersal by Animals (Zoochory)

Birds and mammals, including human beings play an efficient and important role in the dispersal of fruit and seeds. They have the following devices.

i. Hooked fruit: The surface of the fruit or seeds have hooks,(*Xanthium*), barbs (*Andropogon*), spines (*Aristida*) by means of which they adhere to the body of animals or clothes of human beings and get dispersed.

ii. Sticky fruits and seeds:

a. Some fruits have sticky glandular hairs by which they adhere to the fur of grazing animals. Example: *Boerhaavi*a and *Cleome*.

b. Some fruits have viscid layer which adhere to the beak of the bird which eat them and when they rub them on to the branch of the tree, they disperse and germinate. Example: *Cordia* and *Alangium*

iii. Fleshy fruits: Some fleshy fruits with conspicuous colours are dispersed by human beings to distant places after consumption. Example: Mango and *Diplocyclos*



Figure 6.32: Sunflower



Figure 6.33: Papaya

6.4.3 Dispersal by Explosive Mechanism (Autochory)

Some fruits burst suddenly with a force enabling to throw seeds to a little distance away from the plant. Autochory shows the following adaptations.

- Mere touch of some plants causes the ripened fruit to explode suddenly and seeds are thrown out with great force. Example: *Impatiens* (Balsam), *Hura*.
- Some fruits when they come in contact with water particularly after a shower of rain, burst suddenly with a noise and scatter the seeds.Examples: *Ruellia* and *Crossandra.*
- Certain long pods explode with a loud noise like cracker, scattering the seeds in all directions. Example: *Bauhinia vahlii* (Camel's foot climber)
- As the fruit matures, tissues around seeds are converted into a mucilaginous fluid, due to which a **high turgor pressure** develops inside the fruit which leads to the dispersal of seeds.

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